

## **NORMALIZATION EXAMPLE**

## Not in 1NF

| emp_id | emp_name | emp_address | emp_mobile               |
|--------|----------|-------------|--------------------------|
| 101    | Herschel | New Delhi   | 8912312390               |
| 102    | Jon      | Kanpur      | 8812121212<br>9900012222 |
| 103    | Ron      | Chennai     | 7778881212               |
| 104    | Lester   | Bangalore   | 9990000123<br>8123450987 |

# 1NF

| emp_id | emp_name | emp_address | emp_mobile |
|--------|----------|-------------|------------|
| 101    | Herschel | New Delhi   | 8912312390 |
| 102    | Jon      | Kanpur      | 8812121212 |
| 102    | Jon      | Kanpur      | 9900012222 |
| 103    | Ron      | Chennai     | 7778881212 |
| 104    | Lester   | Bangalore   | 9990000123 |
| 104    | Lester   | Bangalore   | 8123450987 |

**Candidate Keys:** {teacher\_id, subject}  
**Non prime attribute:** teacher\_age

| teacher_id | subject   | teacher_age |
|------------|-----------|-------------|
| 111        | Maths     | 38          |
| 111        | Physics   | 38          |
| 222        | Biology   | 38          |
| 333        | Physics   | 40          |
| 333        | Chemistry | 40          |

# 2NF-Full functional dependency

| teacher_id | teacher_age |
|------------|-------------|
| 111        | 38          |
| 222        | 38          |
| 333        | 40          |

**teacher\_details table:**

| teacher_id | subject   |
|------------|-----------|
| 111        | Maths     |
| 111        | Physics   |
| 222        | Biology   |
| 333        | Physics   |
| 333        | Chemistry |

**teacher\_subject table:**

# Third Normal form (3NF)-Transitive dependency

| emp_id | emp_name | emp_zip | emp_state | emp_city | emp_district |
|--------|----------|---------|-----------|----------|--------------|
| 1001   | John     | 282005  | UP        | Agra     | Dayal Bagh   |
| 1002   | Ajeet    | 222008  | TN        | Chennai  | M-City       |
| 1006   | Lora     | 282007  | TN        | Chennai  | Urrapakkam   |
| 1101   | Lilly    | 292008  | UK        | Pauri    | Bhagwan      |
| 1201   | Steve    | 222999  | MP        | Gwalior  | Ratan        |

**Candidate Keys:** {emp\_id}

**Non-prime attributes:** all attributes except emp\_id are non-prime as they are not part of any candidate keys.

Here, emp\_state, emp\_city & emp\_district dependent on emp\_zip.

And, emp\_zip is dependent on emp\_id that makes non-prime attributes (emp\_state, emp\_city & emp\_district) transitively dependent on super key (emp\_id). This violates the rule of 3NF.

| emp_id | emp_name | emp_zip |
|--------|----------|---------|
| 1001   | John     | 282005  |
| 1002   | Ajeet    | 222008  |
| 1006   | Lora     | 282007  |
| 1101   | Lilly    | 292008  |
| 1201   | Steve    | 222999  |

**employee table:**

| emp_zip | emp_state | emp_city | emp_district |
|---------|-----------|----------|--------------|
| 282005  | UP        | Agra     | Dayal Bagh   |
| 222008  | TN        | Chennai  | M-City       |
| 282007  | TN        | Chennai  | Urrapakka m  |
| 292008  | UK        | Pauri    | Bhagwan      |
| 222999  | MP        | Gwalior  | Ratan        |

**Emp\_zip:**

# Boyce Codd normal form (BCNF)

It is an advance version of 3NF

| emp_id | emp_nationality | emp_dept                     | dept_type | dept_no_of_emp |
|--------|-----------------|------------------------------|-----------|----------------|
| 1001   | Austrian        | Production and planning      | D001      | 200            |
| 1001   | Austrian        | stores                       | D001      | 250            |
| 1002   | American        | design and technical support | D134      | 100            |
| 1002   | American        | Purchasing department        | D134      | 600            |

## Functional dependencies in the table above:

emp\_id → emp\_nationality  
emp\_dept → {dept\_type, dept\_no\_of\_emp}

**Candidate key:** {emp\_id, emp\_dept}

The table is not in BCNF as neither emp\_id nor emp\_dept alone are keys.



| emp_id | emp_nationality |
|--------|-----------------|
| 1001   | Austrian        |
| 1002   | American        |

emp\_nationality table:

| emp_id | emp_dept                     |
|--------|------------------------------|
| 1001   | Production and planning      |
| 1001   | stores                       |
| 1002   | design and technical support |
| 1002   | Purchasing department        |

emp\_dept\_mapping table:

| emp_dept                     | dept_type | dept_no_of_emp |
|------------------------------|-----------|----------------|
| Production and planning      | D001      | 200            |
| stores                       | D001      | 250            |
| design and technical support | D134      | 100            |
| Purchasing department        | D134      | 600            |

emp\_dept table:

- **Functional dependencies:**

emp\_id -> emp\_nationality

emp\_dept -> {dept\_type, dept\_no\_of\_emp}

- **Candidate keys:**

For first table: emp\_id

For second table: emp\_dept

For third table: {emp\_id, emp\_dept}

# Sales Records

| Cust ID  | Cust Name   | Item                 | Shipping Address | Newsletter         | Supplier        | Supplier Phone                    | Price |
|----------|-------------|----------------------|------------------|--------------------|-----------------|-----------------------------------|-------|
| at_smith | Alan Smith  | Xbox One             | 35 Palm St.      | Xbox News          | Microsoft       | (800) Buy-XBOX                    | 250   |
| roger25  | Roger Banks | PS4                  | 47 Camp Rd.      | PS News            | Sony            | (800) Buy-SONY                    | 300   |
| wilson44 | Evan Wilson | Xbox One<br>PS Vista | 28 Rock Av.      | Xbox News, PS News | Microsoft, Sony | (800) Buy-XBOX,<br>(800) Buy-SONY | 550   |
| am_smith | Alan Smith  | PS4                  | 47 Camp Rd.      | PS News            | Sony            | (800) Buy-SONY                    | 300   |

# 1NF

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|----------|-------------|----------|------------------|------------|-----------|----------------|-------|
| at_smith | Alan Smith  | Xbox One | 35 Palm St.      | Xbox News  | Microsoft | (800) Buy-XBOX | 250   |
| roger25  | Roger Banks | PS4      | 47 Camp Rd.      | PS News    | Sony      | (800) Buy-SONY | 300   |
| wilson44 | Evan Wilson | Xbox One | 28 Rock Av.      | Xbox News  | Microsoft | (800) Buy-XBOX | 250   |
| wilson44 | Evan Wilson | PS Vista | 28 Rock Av.      | PS News    | Sony      | (800) Buy-SONY | 300   |
| am_smith | Alan Smith  | PS4      | 47 Camp Rd.      | PS News    | Sony      | (800) Buy-SONY | 300   |

# 2NF

| Cust ID  | Cust Name   | Shipping Address | Newsletter | Item     |
|----------|-------------|------------------|------------|----------|
| at_smith | Alan Smith  | 35 Palm St.      | Xbox News  | Xbox One |
| roger25  | Roger Banks | 47 Camp Rd.      | PS News    | PS4      |
| wilson44 | Evan Wilson | 28 Rock Av.      | Xbox News  | Xbox One |
| wilson44 | Evan Wilson | 28 Rock Av.      | PS News    | PS Vista |
| am_smith | Alan Smith  | 47 Camp Rd.      | PS News    | PS4      |

| Item     | Supplier  | Supplier Phone | Price |
|----------|-----------|----------------|-------|
| Xbox One | Microsoft | (800) Buy-XBOX | 250   |
| PS4      | Sony      | (800) Buy-SONY | 300   |
| PS Vista | Sony      | (800) Buy-SONY | 300   |

# 3NF

| Cust ID  | Cust Name   | Shipping Address | Newsletter | Item     |
|----------|-------------|------------------|------------|----------|
| at_smith | Alan Smith  | 35 Palm St.      | Xbox News  | Xbox One |
| roger25  | Roger Banks | 47 Camp Rd.      | PS News    | PS4      |
| wilson44 | Evan Wilson | 28 Rock Av.      | Xbox News  | Xbox One |
| wilson44 | Evan Wilson | 28 Rock Av.      | PS News    | PS Vista |
| am_smith | Alan Smith  | 47 Camp Rd.      | PS News    | PS4      |

| Item     | Supplier  | Price |
|----------|-----------|-------|
| Xbox One | Microsoft | 250   |
| PS4      | Sony      | 300   |
| PS Vista | Sony      | 300   |

| Supplier  | Supplier Phone |
|-----------|----------------|
| Microsoft | (800) Buy-XBOX |
| Sony      | (800) Buy-SONY |